

# Classical Demand I: UMP and EMP

Advanced Microeconomics 1: Economic Decision Making. Lecture 2

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## Where we left the question

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- **Done, with no assumptions:** household inflation  $\pi^h = \sum_I b_I^h \pi_I$  — who faced what.
- **Done, with one assumption:** basket indexation over-compensates — three lines of revealed preference.
- **Missing:** the *right* compensation. “As well off as before” still has no number attached.

Today we cheat, deliberately: **assume the household's ranking is known**, and construct the ideal answer. Lectures 3–4 earn back what we assume today.

## Today's route

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1. A ranking is not yet arithmetic: when does  $\succsim$  become a **function**  $u$ ?
2. Two optimization problems — one natural (UMP), one inverted (EMP).
3. The inverted one produces **the cost of living as a mathematical object**:  $e(p, u)$ .
4. From  $e$ : the ideal answers to the 2022 question — CV and EV, in euros, per household.

Invoice item 3: define the ideal answer. Every object below exists for that reason.

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## **From ranking to function**

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## Standing assumptions on $\succsim$ , and why each is bought

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### Definition · monotone / locally nonsatiated

$\succsim$  is **monotone** if  $y \gg x \Rightarrow y \succ x$ ; **strongly monotone** if  $y \geq x$ ,  $y \neq x \Rightarrow y \succ x$ . It is **locally nonsatiated (LNS)** if for every  $x \in X$  and every  $\varepsilon > 0$  there is  $y \in X$  with  $\|y - x\| \leq \varepsilon$  and  $y \succ x$ .

- Monotone: more is better — plausible for broad aggregates. LNS is weaker: no bliss point nearby.
- Why bought: **LNS is exactly what makes households spend their budget** — Walras' law, an assumption about data in Lecture 1, becomes a theorem today.

# Convexity

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## Definition · convex preferences

$\succsim$  is **convex** if every upper contour set  $\{y \in X : y \succsim x\}$  is convex:  $y \succsim x$  and  $z \succsim x$  imply  $\alpha y + (1 - \alpha)z \succsim x$  for  $\alpha \in [0, 1]$ . **Strictly convex**: the mixture is  $\succ x$  whenever  $y \neq z$ ,  $\alpha \in (0, 1)$ .

- Reads as: a taste for diversification; diminishing marginal rates of substitution.
- Why bought: convexity makes the set of optimal choices convex; **strict convexity makes demand a function** — one bundle per budget, which is what data look like.

## Why a utility function can fail to exist

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Rationality alone is not enough. The classic counterexample, on  $X = \mathbb{R}_+^2$ :

### Definition · lexicographic preferences

$x \succsim y$  if either  $x_1 > y_1$ , or  $x_1 = y_1$  and  $x_2 \geq y_2$ .

- Complete, transitive, strongly monotone, strictly convex — everything so far. Yet **no utility function represents it**.
- Reason: no two distinct bundles are indifferent, so every indifference set is a single point. A two-dimensional continuum of indifference sets must each receive its own real number, in order — and there are *not enough real numbers*.

## Continuity, and Debreu's theorem

### Definition · continuity

$\succsim$  is **continuous** if for any sequence  $\{(x^n, y^n)\}_{n=1}^{\infty}$  with  $x^n \succsim y^n$  for all  $n$ ,  $x^n \rightarrow x$  and  $y^n \rightarrow y$ , we have  $x \succsim y$ . (Preferences cannot jump at the limit.)

- Lexicographic fails it:  $x^n = (\frac{1}{n}, 0) \succsim y^n = (0, 1)$  for every  $n$ , yet in the limit  $y = (0, 1) \succ (0, 0) = x$ .

### Theorem · Debreu

If the rational preference relation  $\succsim$  on  $X$  is continuous, there is a *continuous* utility function  $u(x)$  that represents  $\succsim$ .

Proof: beyond our scope; interested students are referred to MWG (as in Frederick's slides). With Lecture 1's easy direction: rational + continuous  $\Rightarrow$  representable, and representable  $\Rightarrow$  rational.



## Ordinal, not cardinal

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- $u$  is not unique: any strictly increasing  $f$  makes  $v = f(u)$  represent the *same*  $\succsim$ .  
E.g.  $x_1^\alpha x_2^{1-\alpha}$  and  $\alpha \ln x_1 + (1 - \alpha) \ln x_2$ : one preference, two functions.
- The dictionary: monotone  $\succsim \Rightarrow u$  increasing; (strictly) convex  $\succsim \Rightarrow u$  (strictly) **quasiconcave**:  $\{y : u(y) \geq u(x)\}$  convex. Both are *ordinal* — they survive increasing transforms.
- **Levels of  $u$  mean nothing.** “Utility fell by 3” is not a sentence. This is precisely why the inflation question cannot stop at  $u$ : the answer must be denominated in something real. Euros, by the end of today.

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## **The consumer's problem**

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## The utility maximization problem

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Rational, continuous, LNS  $\succsim$  on  $X = \mathbb{R}_+^L$ , represented by continuous  $u$ ; prices  $p \gg 0$ , wealth  $w > 0$ :

$$(UMP) \quad \max_{x \geq 0} u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

### Theorem · existence

If  $u(\cdot)$  is continuous, the UMP has a solution.

► proof

Two objects fall out, and the rest of the block lives off them: the **maximizer**  $x(p, w)$  and the **value**  $v(p, w)$ .

## Walrasian demand: Lecture 1's laws become theorems

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### Theorem · MWG 3.D.2

If  $u$  is a continuous utility function representing an LNS preference relation  $\succsim$  on  $\mathbb{R}_+^L$ , then  $x(p, w)$  satisfies

- (i) **homogeneity of degree zero:**  $x(\alpha p, \alpha w) = x(p, w)$  for  $\alpha > 0$ ;
- (ii) **Walras' law:**  $p \cdot x = w$  for all  $x \in x(p, w)$ ;
- (iii) **convexity/uniqueness:** convex  $\succsim \Rightarrow x(p, w)$  a convex set; strictly convex  $\succsim \Rightarrow$  single-valued.

► proof

In Lecture 1, (i) and (ii) were *assumptions* about data. Under today's hypothesis they are **consequences** — the first repayment of the loan.

## Solving it: Kuhn–Tucker

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If  $u$  is continuously differentiable, an optimum  $x^*$  admits a multiplier  $\lambda \geq 0$  with, for each  $\ell$ :

$$\frac{\partial u(x^*)}{\partial x_\ell} \leq \lambda p_\ell, \quad \text{with equality if } x_\ell^* > 0.$$

- Interior optimum ( $x^* \gg 0$ ): for any two goods,  $\frac{\partial u(x^*)/\partial x_\ell}{\partial u(x^*)/\partial x_k} = \frac{p_\ell}{p_k}$  — **MRS = price ratio**. Only relative prices appear: h.d.0, visible in the first-order condition.
- $\lambda$  is the shadow value of the budget:  $\nabla u \cdot D_w x = \lambda p \cdot D_w x = \lambda$  — the marginal utility of one more euro.
- The conditions are also sufficient if  $u$  is quasiconcave and monotone with  $\nabla u(x) \neq 0$ .

## Worked: Cobb–Douglas

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$u(x) = x_1^\alpha x_2^{1-\alpha}$ ,  $\alpha \in (0, 1)$ . Interior FOCs + Walras' law give

$$x_1(p, w) = \frac{\alpha w}{p_1}, \quad x_2(p, w) = \frac{(1 - \alpha) w}{p_2}.$$

- **Constant budget shares:**  $b_1 = \alpha$ ,  $b_2 = 1 - \alpha$  — exactly the  $b_i^h$  of the motivation charts. A Cobb–Douglas household *is* a row of the NBB quartile table.
- Small enough to carry through the block: the ideal answer is computed for this household before the hour ends.
- EC1: the subsistence version  $u = (x_1 - s_1)^\alpha (x_2 - s_2)^{1-\alpha}$  — shares no longer constant; the poor pinned to necessities.

## The indirect utility function

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### Definition · indirect utility

$v(p, w) = u(x^*)$  for any  $x^* \in x(p, w)$ : the maximal utility attainable at  $(p, w)$ .

### Proposition · properties (MWG 3.D.3)

$v(p, w)$  is (i) h.d.0; (ii) increasing in  $w$  and nonincreasing in  $p_\ell$ ; (iii) quasiconvex:  $\{(p, w) : v(p, w) \leq \bar{v}\}$  is convex; (iv) continuous.

Proof deferred to MWG, as in Frederick's slides. A fifth property — Roy's identity — is Lecture 3's engine.

“Worse off from the price change” now has a formal reading:  $v(p^1, w) < v(p^0, w)$ . A *direction*, at last — but not yet a *size*.

## Why $v$ cannot be the final answer

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- $v$  depends on which representation we picked: replace  $u$  by  $f(u)$  and every value of  $v$  changes. The size of  $v(p^1, w) - v(p^0, w)$  is meaningless.
- Belgium does not index wages in utils.
- The fix is to change the question. Instead of “how much utility from this budget?”, invert: “how much budget for this standard of living?” Money in, money out.



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**The answer's object: the expenditure function**

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## The expenditure minimization problem

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For  $p \gg 0$  and  $u > u(0)$ :

$$(\text{EMP}) \quad \min_{x \geq 0} p \cdot x \quad \text{s.t.} \quad u(x) \geq u.$$

### Definition · expenditure function

$e(p, u) = p \cdot x^*$  for any solution  $x^*$ : the *minimal cost of reaching the standard of living  $u$  at prices  $p$ .*

- Read it again with 2022 in mind:  $e(p^{2022}, u^{2021})$  is **what it costs to be as well off as before** — the phrase the course has been chasing, now a mathematical object.
- The *dual* of the UMP: objective and constraint swap roles.

## UMP and EMP are the same problem, run backwards

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### Theorem · MWG 3.E.1

Under the maintained assumptions (continuity, LNS), for  $p \gg 0$ :

- (i)  $x^*$  optimal in the UMP at wealth  $w > 0 \Rightarrow x^*$  optimal in the EMP at required utility  $u(x^*)$ ; the minimized expenditure is exactly  $w$ .
- (ii)  $x^*$  optimal in the EMP at  $u > u(0) \Rightarrow x^*$  optimal in the UMP at wealth  $p \cdot x^*$ ; the maximized utility is exactly  $u$ .

► proof

Hence the two value functions are inverses of each other in their last argument:

$$e(p, v(p, w)) \equiv w, \quad v(p, e(p, u)) \equiv u.$$

## Properties of the cost of living

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### Theorem · MWG 3.E.2

$e(p, u)$  is

- (i) homogeneous of degree **one** in  $p$ ;
- (ii) strictly increasing in  $u$  and nondecreasing in each  $p_\ell$ ;
- (iii) **concave** in  $p$ ;
- (iv) continuous in  $p$ , for  $p \gg 0$ .

► proof

- (i): double every price and the cost of the old life doubles — the indexation instinct, exactly right *for proportional* inflation. 2022 was not proportional.
- (iii) is the sleeper. Concavity: as one price rises, cost rises *less than linearly*, because the household substitutes. **Lecture 1's over-compensation theorem is the shadow of this curvature.** Lecture 3 measures it.

## Hicksian demand

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### Definition · Hicksian (compensated) demand

$h(p, u) \subseteq \mathbb{R}_+^L$ : the solution set of the EMP — the cheapest bundle(s) delivering the standard of living  $u$  at prices  $p$ .

### Theorem · MWG 3.E.3

For  $p \gg 0$ : (i) **h.d.0 in  $p$** :  $h(\alpha p, u) = h(p, u)$ ; (ii) **no excess utility**:  $u(x) = u$  for any  $x \in h(p, u)$ ; (iii)  $\text{convex} \succsim \Rightarrow h(p, u) \text{ convex}$ ;  $\text{strictly convex} \Rightarrow \text{single-valued}$ . [▶ proof](#)

Duality identities, from the equivalence theorem:

$$h(p, u) \equiv x(p, e(p, u)), \quad x(p, w) \equiv h(p, v(p, w)).$$

Not directly observable — no dataset holds utility fixed. Its road to the data is Lecture 3.

From  $(p^0, w)$  with  $x^0 = x(p^0, w)$ , prices move to  $p^1$ . Two ways to “compensate”:

### Slutsky (Lecture 1)

$\Delta w_S = p^1 \cdot x^0 - w$ : keep the *old bundle* affordable.

Observable today. **Over-pays** (the L1 theorem). This is what basket indexation does.

### Hicksian (today)

$\Delta w_H = e(p^1, u^0) - w$ : keep the *old standard of living* reachable.

**Exact** — the ideal answer. Needs preferences.

$$\Delta w_H \leq \Delta w_S \quad \text{since } e(p^1, u^0) \leq p^1 \cdot x^0 \text{ (} x^0 \text{ attains } u^0 \text{, so it is feasible in the EMP).}$$

The gap between the columns is the **substitution bias** — the euro amount at stake in the indexation debate.

## The compensated law of demand, from preferences

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### Theorem · MWG 3.E.4

Under the maintained assumptions, with  $h(p, u)$  single-valued for  $p \gg 0$ : for all  $p'$  and  $p''$ ,

$$(p'' - p') \cdot [h(p'', u) - h(p', u)] \leq 0.$$

► proof

- Own-price version: compensated own-price effects are nonpositive, globally. *Marshallian* demand obeys no such law (Giffen); holding the standard of living fixed is what buys the sign.
- Lecture 1 proved this law from WARP, for *Slutsky*-compensated changes. Same law, second derivation, deeper hypothesis. The two constructions collide — productively — in Lecture 3.

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**The ideal answer, stated**

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## Money-metric utility: euros as the unit

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Fix any reference prices  $\bar{p} \gg 0$  and consider, as a function of  $(p, w)$ :

$$e(\bar{p}, v(p, w)).$$

- This is itself an indirect utility function — a strictly increasing transform of  $v$  — but its unit is **euros at prices  $\bar{p}$** : the budget needed, at reference prices, to live as well as you do at  $(p, w)$ .
- Differences of *this* object are meaningful money amounts. The representation-dependence of  $v$  is gone.
- One free choice remains: *which*  $\bar{p}$ . The two natural candidates are the old prices and the new — and each choice has a name.

Let  $u^0 = v(p^0, w)$ ,  $u^1 = v(p^1, w)$ ; note  $e(p^0, u^0) = e(p^1, u^1) = w$ .

### Definition · equivalent variation

$EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0) = e(p^0, u^1) - w$ : the euro amount, *at old prices*, equivalent to the price change.

### Definition · compensating variation

$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0) = w - e(p^1, u^0)$ : minus the payment that would exactly restore the old standard of living *at new prices*.

- For 2022:  $-CV^h = e^h(p^1, u^0) - w$  is the correct indexation payment for household  $h$  — the number the health index approximates with one basket and one average household.
- Both are ideal; they differ in the reference. Which one policy owes is a genuine choice — it returns, with teeth, in Lecture 4.

### Definition · Konüs (true cost-of-living) index

$$P_K(p^0 \rightarrow p^1; u^0) = \frac{e(p^1, u^0)}{e(p^0, u^0)} \quad \text{vs. Laspeyres:} \quad P_L = \frac{p^1 \cdot x^0}{p^0 \cdot x^0}.$$

### Proposition · Laspeyres bounds the truth from above

$P_K \leq P_L$ . *Proof:*  $x^0$  attains  $u^0$ , so  $e(p^1, u^0) \leq p^1 \cdot x^0$ ; and  $e(p^0, u^0) = p^0 \cdot x^0 = w$ .  $\square$

- Two lines — Lecture 1's three-line theorem, rewritten in the answer's own language. Fixed-basket indices (the CPI family, the health index) are Laspeyres-type: **they mechanically over-state the rise in the cost of living.**
- By how much? A second-order question — literally. Lecture 3.

Both objects, computed on UK scanner data through the 2021–23 surge: Chen, Levell & O'Connell (2025) — their “quality-of-living index” is exactly the proportional CV of the last slide. §3 is this lecture's notation in the field; Lectures 3–4 use their numbers.

Cobb–Douglas household, goods = (energy, other), share  $\alpha$  on energy. Then

$$e(p, u) = u \left( \frac{p_E}{\alpha} \right)^\alpha \left( \frac{p_O}{1-\alpha} \right)^{1-\alpha} \implies \frac{e(p^1, u^0)}{w} = \left( \frac{p_E^1}{p_E^0} \right)^\alpha \quad (\text{only } p_E \text{ moves}).$$

Energy prices rise 50%. Two households from the motivation chart:

|                       | exact: $-CV/w = 1.5^\alpha - 1$ | basket: $\Delta w_S/w = \alpha \times 50\%$ | over-pay |
|-----------------------|---------------------------------|---------------------------------------------|----------|
| poor, $\alpha = 0.10$ | 4.1%                            | 5.0%                                        | 0.9 pp   |
| rich, $\alpha = 0.06$ | 2.5%                            | 3.0%                                        | 0.5 pp   |

Both L1 results, now with numbers: the *right* payments differ across households (4.1 vs 2.5), and the basket over-pays both. One index, one number, misses both facts.

## Where the question stands

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### Takeaways

- **The ideal answer exists:**  $e(p^1, u^0) - w$ , per household — “as well off as before” is now arithmetic. EV and CV name the two natural references.
- **Two accounting laws upgraded:** h.d.0 and Walras’ law are theorems under the maintained hypothesis, not assumptions about data.
- **The catch, stated plainly:** everything today was built from  $u$  — which no dataset contains. The answer is ideal and, so far, *uncomputable*.
- **Next:** Lecture 3 connects  $e$ ,  $h$  and  $v$  to observable demand. Four bridges; all of them derivatives.

**Required:** MWG 3.A–3.E.

**Papers:** Hausman (AER 1981), §I–II, before Lecture 4; Chen–Levell–O’Connell (IFS 2025), §3 — tonight’s notation, in the field.

**Ahead:** the practical is Thursday — laptop, working R installation, your quiz ID.

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## **Appendix: proofs**

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## Proof: the UMP has a solution

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- The budget set  $B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$  is **bounded**: for each  $\ell$ ,  $x_\ell \leq w/p_\ell$  for every  $x \in B_{p,w}$  (this uses  $p \gg 0$ ).
- It is **closed**: an intersection of closed half-spaces with the closed orthant.
- Closed and bounded in  $\mathbb{R}^L$  = **compact**; and  $B_{p,w} \neq \emptyset$  since  $0 \in B_{p,w}$ . A continuous function attains a maximum on a nonempty compact set. ■

Where it can fail: some  $p_\ell = 0$  (budget set unbounded), or  $u$  discontinuous — lexicographic preferences again. Another job continuity does.

## Proof: properties of Walrasian demand (MWG 3.D.2)

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- (i) *h.d.o.*  $B_{\alpha p, \alpha w} = \{x : \alpha p \cdot x \leq \alpha w\} = B_{p, w}$ : same constraint set, same objective, hence the same maximizers.
- (ii) *Walras' law*. Suppose  $x \in x(p, w)$  with  $p \cdot x < w$ . Then a small ball around  $x$  lies inside the budget set, and LNS supplies  $y$  in that ball with  $y \succ x$ : an affordable strict improvement, contradicting optimality. So  $p \cdot x = w$ .
- (iii) *Convexity/uniqueness*. Let  $x, x' \in x(p, w)$ , both with utility  $u^*$ . The mixture  $x'' = \alpha x + (1 - \alpha)x'$  is affordable ( $B_{p, w}$  convex) and quasiconcavity gives  $u(x'') \geq u^*$ , so  $x'' \in x(p, w)$ : the solution set is convex. If  $u$  is *strictly* quasiconcave and  $x \neq x'$ , then  $u(x'') > u^*$  — contradicting maximality of  $u^*$ . So the solution is unique. ■



## Proof: UMP $\iff$ EMP (MWG 3.E.1)

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(i) Let  $x^*$  solve the UMP at wealth  $w$ ; by Walras' law,  $p \cdot x^* = w$ . Suppose  $x^*$  does *not* solve the EMP at utility  $u(x^*)$ : some  $x'$  has  $u(x') \geq u(x^*)$  and  $p \cdot x' < w$ . By LNS there is  $x''$  near  $x'$  with  $u(x'') > u(x') \geq u(x^*)$  and still  $p \cdot x'' < w$  — an affordable bundle beating  $x^*$  in the UMP. Contradiction. The minimized expenditure is  $p \cdot x^* = w$ .

(ii) Let  $x^*$  solve the EMP at  $u > u(0)$ ; then  $x^* \neq 0$ , so  $w' := p \cdot x^* > 0$ . Suppose  $x^*$  does *not* solve the UMP at  $w'$ : some  $x'$  has  $p \cdot x' \leq w'$  and  $u(x') > u(x^*) \geq u$ . By continuity,  $\alpha x'$  with  $\alpha < 1$  close to 1 still has  $u(\alpha x') > u$ , at cost  $\alpha p \cdot x' < w'$  — cheaper and feasible in the EMP, contradicting minimality. Finally,  $u(x^*) = u$  exactly: if  $u(x^*) > u$ , the same scaling applied to  $x^*$  itself yields a cheaper feasible bundle. ■

## Proof: properties of the expenditure function (MWG 3.E.2)

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- (i) *h.d. 1.* The constraint set  $\{x : u(x) \geq u\}$  does not involve  $p$ ; scaling  $p \rightarrow \alpha p$  scales the objective, hence the minimized value, by  $\alpha$ .
- (ii) *Strictly increasing in  $u$ .* Suppose  $u' > u$  but  $e(p, u') \leq e(p, u)$ , and let  $x'$  attain  $u'$  at cost  $e(p, u')$ . By continuity,  $\alpha x'$  with  $\alpha < 1$  near 1 still gives utility  $> u$ , at cost  $\alpha e(p, u') < e(p, u)$ : contradicts the definition of  $e(p, u)$ . (Nondecreasing in  $p_\ell$ : raising  $p_\ell$  weakly raises the cost of every feasible bundle.)
- (iii) *Concavity.* Fix  $\bar{u}$ ; let  $p'' = \alpha p + (1 - \alpha)p'$  and let  $x''$  be optimal at  $p''$ . Then  $x''$  attains  $\bar{u}$ , hence is *feasible* (not necessarily optimal) at prices  $p$  and at prices  $p'$ :

$$e(p'', \bar{u}) = p'' \cdot x'' = \alpha p \cdot x'' + (1 - \alpha) p' \cdot x'' \geq \alpha e(p, \bar{u}) + (1 - \alpha) e(p', \bar{u}). \quad \blacksquare$$

(iv) Continuity: the maximum theorem; see MWG.

## Proof: properties of Hicksian demand (MWG 3.E.3)

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- (i) *h.d.0 in p*. Minimizing  $\alpha p \cdot x$  and minimizing  $p \cdot x$  over the same constraint set  $\{x : u(x) \geq u\}$  have the same solutions.
- (ii) *No excess utility*. Suppose  $x \in h(p, u)$  with  $u(x) > u$ . Since  $u > u(0)$ ,  $x \neq 0$ , so  $p \cdot x > 0$ . By continuity,  $\alpha x$  with  $\alpha < 1$  near 1 still has  $u(\alpha x) > u$  — feasible and strictly cheaper. Contradiction; so  $u(x) = u$ .
- (iii) *Convexity/uniqueness*. The constraint set  $\{x : u(x) \geq u\}$  is convex when  $u$  is quasiconcave, and the objective is linear: two minimizers cost the same, and any mixture of them is feasible and costs the same — the solution set is convex. Under *strict* convexity, the mixture of two distinct minimizers lies strictly inside the upper contour set, so a small radial contraction stays feasible and cuts cost — forcing uniqueness. ■

## Proof: the compensated law of demand (MWG 3.E.4)

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Fix  $u$ . Each of  $h(p', u)$  and  $h(p'', u)$  attains utility  $u$ , so each is *feasible* in the other's problem. Cost-minimality therefore gives two inequalities:

$$\begin{aligned} p'' \cdot h(p'', u) &\leq p'' \cdot h(p', u) && (h(p', u) \text{ feasible at prices } p''), \\ p' \cdot h(p', u) &\leq p' \cdot h(p'', u) && (h(p'', u) \text{ feasible at prices } p'). \end{aligned}$$

Add the first to minus the second:

$$(p'' - p') \cdot [h(p'', u) - h(p', u)] \leq 0. \blacksquare$$

Two lines of feasibility — no calculus, no WARP. Compare Lecture 1: the same law from choice consistency needed the compensation construction. From preferences it is nearly free.